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Effects of Financial Constraints on Supply Chain Financing Choices and Operational Decisions

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Abstract. Bank credit access allows firms to borrow funds from banks and other financial institutions. Although theories have been developed on how bank financing affects firms' operational decisions, empirical investigations in operations management (OM) literature remain limited. This study examines firms' inventory management strategies in response to the enhanced credit lines from the adoption of interstate bank branching laws, which introduced staggered access to bank credit for operational firms. By merging multiple data-sets with credit line information from 10-K SEC filings, we constructed a panel data set covering 1990–2005. Utilizing a difference-in-differences (DID) approach, we find that improved bank credit access leads to a 6% faster inventory turnover, rather than an increase in inventory investment. This outcome appears to stem from short-term increases in capacity investments leveraging their enhanced bank credit and increased use of trade credit, alongside long-term improvements in infrastructure and capital intensity. While these developments are more pronounced for small firms in concentrated markets, their advantage translates into increased competition, negatively impacting the market position and profitability of larger firms. Additionally, in a broader supply chain context, focal firms experience faster inventory turnover when their major customers access more bank credit. The effect is especially prominent for large suppliers in competitive markets, driven by increased sales volumes. This research enhances understanding of the extensive impact of bank credit on operational management and offers insights for policymakers about the diverse effects of bank regulations across different market players and sectors.

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1. Introduction

Bank credit accessibility, referring to a firm's ability to secure financing from banks and other financial institutions, is crucial for businesses across various industrial sectors and stages of development. Adequate bank credit is essential for the long-term success of a company, as it facilitates acquisition of updated equipment and adoption of new technologies (McCardle 1985, Erat and Kavadias 2006), both of which can enhance productivity, efficiency, and competitiveness. Bank credit also plays a critical role in the short term, for maintaining sufficient working capital to cover immediate expenses and effectively manage cash-flow fluctuations (Chen and Kieschnick 2018) in order to support day-to-day operations.

An important daily operational decision that is closely linked to bank credit access is inventory management. As of fiscal year 2022, it has been observed that inventory represents more than 35%[1] of current assets among U.S. public firms across various industries, including construction, wholesale, retail, and manufacturing. To maintain adequate inventory levels and manage cash-flow fluctuations, many businesses have turned to bank credit as an inventory financing source, for example, inventory loans and inventory lines of credit (Buzacott and Zhang 2004). The relationship between inventory decisions and financial considerations has been explored in recent literature that builds analytical models on trade credit and inventory decisions (e.g., Yang and Birge 2018), inventory management with financial constraints (e.g., Chao et al. 2008), and inventory risk sharing in supply chains (Lai et al. 2009). While many analytical models have been established for making simultaneous financing and operational decisions, limited empirical examinations have been done to examine the interplay between these decisions. A few empirical works (e.g., Wu et al. 2019, Chen et al. 2023) in the OM-finance interface focus on relationship between inventory and its financing channel, either bank credit or trade credit, of firms. Continuing with this line of research, this study considers multiple financing channels and operational factors to empirically investigate the causal impact of bank credit accessibility on inventory decisions as well as the mechanism through which this impact manifests.

The relationship between bank credit access and inventory is complex and multifaceted. It is not entirely clear how a firm will adjust its inventory decision when it has greater access to bank credit. On the one hand, firms with greater access to credit may be more willing to maintain higher inventory levels to meet customer demand, take advantage of bulk purchase discounts, and mitigate supply chain risks (Tomlin 2006). On the other hand, an aggregate-planning-based view suggests that firms with convenient access to bank credit may make investments to balance the capacity-inventory trade-off, which enables those firms to enhance overall operational efficiency and increase inventory turnover (Mellichamp and Love 1978, Ciarallo et al. 1994, Vollmann et al. 2004).

In the broader context of supply chain management, the effect of a firm's increase in bank credit can ripple through the supply chain. Regarding the flow of materials, the adjusted inventory decisions made by the focal firm directly influence the demand or supply patterns of its supply chain partners, prompting them to adjust their own inventory decisions (Krishnan and Winter 2010). Furthermore, a focal firm's bank credit can affect its financial interdependence with its supply chain partners (Casey and O'Toole 2014, Shenoy and Williams 2017) potentially influencing the cash and material flow dynamics across the supply chain.

The complex interplay between financial aspects and operational choices, coupled with the influence of market- and firm-specific attributes, has not been sufficiently probed existing empirical literature of operations management and finance. This paper aims to bridge this gap by addressing the following research questions: How does a firm's access to bank credit influence its inventory turnover? Furthermore, within a broader context of supply chain networks, how do a firm's inventory dynamics interact with the access to bank credit of its competitors and supply chain partners? By addressing these questions, this paper focuses on the special angle of bank credit accessibility, aiming to shed light on the intricate nexus between financial elements and operational decisions when firm-specific characteristics are controlled.

To answer our research questions, it is imperative to take into account a principal empirical challenge that stems from the interdependence between financial and operational decisions. This simultaneity is manifested in scenarios such as joint production and financing decisions (Xu and Birge 2004), inventory management decisions under the influence of trade credit conditions (Haley and Higgins 1973), and firm liquidity constraints (Chao et al. 2008), as well as the incorporation of risk-sharing considerations in supply chain financing (Lai et al. 2009, Kouvelis and Zhao 2012, Yang and Birge 2018). Such intertwined decision-making processes necessitate a causal inference approach in the empirical analysis to accurately discern the relationships and impacts in question.

To address this challenge, this paper relies on exploiting a policy intervention that imparted an exogenous shock to banking liquidity. The Interstate Banking and Branching Efficiency Act (IBBEA) of 1994 allowed commercial banks to establish branches outside their home state without requiring separate charters or capital. This facilitated the entry of out-of-state banks into local banking markets and substantially enhanced firms' access to these banking markets (Johnson and Rice 2008). The IBBEA left each state the option of implementing its provisions under restrictive terms. From 1994 to 2001, states gradually started relaxation of the constraints in a staggered way (Johnson and Rice 2008, Rice and Strahan 2010). The gradual and staggered implementation of the IBBEA generated variations in both cross-sectional and time-series access to local banking markets for firms, allowing us to make causal inferences.

Using this event, we adopt a difference-in-differences (DID) research design consistent with state-of-the-art literature in management and applied economics (e.g., Beck et al. 2010, Chen et al. 2021, Lin et al. 2021) that examines the effects of exogenous policy changes on economic decisions. To estimate the DID model, we employ a stacked regression approach (Cengiz et al. 2019, Slutzky and Zeume 2023, Li 2024), which allows us to isolate the effect of IBBEA deregulation on bank credit provisioning and resultant inventory decisions, while mitigating the potential bias of estimating staggered treatment effects with a two-way fixed effects regression (Baker et al. 2022).

To conduct our empirical analysis, we collate a firm-year panel data for 1990–2005 by collecting firm-specific financial and operational characteristics from multiple sources and augment the data with bank credit information. Since the bank credit information is not readily included in any existing data set for our study period, we acquire this data by text-mining 10K SEC filings, in which firms typically provide details about their credit lines and their unused credit. We integrate the individual firm's information with supply chain information. With a sample of 662 U.S. public firms and 6308 firm-year observations, we analyze the impact of expanded bank credit on firms' operational decisions using both *macro* and *micro* approaches. The *macro* analysis examines the shifts in firm behavior following the major policy change (i.e., the IBBEA deregulation), while the *micro* analysis takes advantage of firm-level credit line information, distinguishing firm-specific credit changes between those firms that experienced a credit increase and those that did not. In the micro approach, we use weighting/matching schemes to balance other covariates between the two groups and control for the confounding effects.

We observe that, on average, inventory turnover increases by 6% after the deregulation, with a notably stronger increase among firms that have expanded their bank credit, even after controlling for potential confounding bias. We investigate aggregate planning theory as a potential mechanism through which these effects could manifest. Our analysis reveals that firms benefiting from bank credit expansion tend to reduce their inventory levels while increasing capital investments in the short term. Over time, this shift leads to greater capital intensity and improved operational efficiency, which further increases inventory turnover in the long run. This effect is particularly evident for small firms operating in concentrated markets, but negatively affects the market position and profitability of larger competitors in these environments. On the customer side, we find that focal firms' inventory turnover also increases when their major customers gain greater access to bank credit. This positive effect is particularly significant for large firms in competitive markets. This increase can be attributed to a higher volume of sales between large firms and their major customers as those customers experience greater access to bank credit. Our findings are robust to alternative explanations and remain consistent across alternative variable and model specifications. Additional heterogeneity checks reveal that less liquid firms and non-commodity industries rely more on bank credit for inventory decisions, further suggesting that credit access impacts firms and industries differently.

The contribution of this paper is threefold. This is the first paper to demonstrate the dynamic impact of credit shocks on inventory and capacity decisions in both the short and long term through the aggregated planning framework. Rather than increasing inventory levels, firms allocate additional financial resources toward expanding production capacity, which in turn affects inventory dynamics over time. In the short term, firms benefiting from expanded bank credit primarily use undrawn credit as a form of insurance. This allows them to secure additional trade credit, together with enhanced bank loans, reduced inventory levels, and lower stocking costs, to facilitate capital investments. In the long term, as a result of these capital investments, these firms achieve more effective inventory management, leading to faster inventory turnover.

Second, this study establishes a causal relationship between enhanced bank credit and inventory decisions and demonstrates the dynamic impact of increased credit on the operational trade-off. Unlike the commonly studied negative financial crisis shocks, our paper examines a period characterized by a sustained positive financial availability shock, offering a unique context for analysis. As a result, the relationships we uncover differ from those found in crisis-focused studies. We show that increased access to credit enhances inventory turnover by enabling firms to expand, increase capital investments, and improve operational efficiency, with bank and trade credits serving as complementary financial resources in this process. This finding advances our understanding in how firms dynamically adjust their operational strategies in response to sustained positive financial shocks, such as policy changes or macroeconomic shifts, contributing uniquely to the broader literature studying financial constraints.

Lastly, due to the lack of detailed credit line information, previous studies on IBBEA deregulation assumed a homogeneous effect of the deregulation on firms access to bank financing. In contrast, our research leverages the most comprehensive data set on credit lines we obtained through text mining. This allows us to identify heterogeneous effects of deregulation on credit lines accurately. This approach enables us to identify which types of firms benefit from increased access to credit and to more precisely assess how credit availability and usage influence firms operational and investment decisions. This data set offers a robust basis for future studies evaluating the polices effectiveness and broader implications, and will be a valuable resource for future research.

2. Literature and Hypotheses

2.1. Related Literature

Our study relates to the literature at the interface of operations management and finance that builds analytical models to study the interaction between financial and operational decisions and how production and operational choices are affected by financial constraints. Xu and Birge (2004) analyze the interactions between a firm's production and financing decisions as a trade-off between the tax benefits of debt and financial-distress costs. The results illustrate greater firm value sensitivity to production decisions than to financing decisions and that low-margin producers face significant risk in not coordinating production and financing decisions. The authors in another paper (Xu and Birge 2008) explore the relationship between operating conditions and financial leverage. They observe that financial leverage can increase as margins reach either low or high values. They provide empirical support of these observations for the nonlinear cubic relationship between profitability and leverage. Chao et al. (2008) consider a classic dynamic inventory control problem and find that the replenishment decisions of the retailer are constrained by cash flow.

Lai et al. (2009) examine the impact of financial constraints on supply chain efficiency. They show that without financial constraints, the supplier always prefers the consignment mode, taking full inventory risk, but in the presence of financial constraints, the supplier will sell part of the inventory to the retailer through preorder, to share the inventory risk.

Our paper is also related to the literature of sources of financing. As specific sources of financing, bank credit and trade credit have been demonstrated to be associated with operational decisions. In the field of the OM-finance interface, studies have explored the supplier's advantage in extending credit to their buyers, relative to financial institutions. Haley and Higgins (1973) is the first study that draws from the economic order quantity (EOQ) model to highlight the importance of coordinating trade credit terms and inventory decisions. More recently, Yang and Birge (2018) and Kouvelis and Zhao (2012) study the inventory financing pattern to understand the risk-sharing role of trade credit. Yang and Birge (2018) also provide empirical evidence of the inventory financing pattern that is consistent with what their model predicts. These studies provide insights that bank and trade credit cannot replace each other in creating value in the supply chain and that each plays an indispensable role in operational decisions. While many analytical models in the OM-finance interface literature simultaneously consider financing and operational decisions, limited empirical examinations have been done. Our paper contributes to this stream of literature by empirically examining the effects of financial constraints on financing choices and operational decisions.

The OM-finance interface literature has recently seen an increasing number of empirical works (e.g., Gaur et al. 2005, Tunca and Zhu 2018). Some works in this literature provide empirical evidence of the interaction between financial and operational decisions of firms. Cunningham (2004) focuses on Canadian manufacturing firms and tests the interaction between cash flow and finance and inventory investment. The study finds that the relationship is nonlinear and heterogeneous among firms with different ages, sizes, and financial constraints. Wu et al. (2019) investigate the effects of bank credit availability, trade credit, and the capital cost on inventory decisions. They find that the primary role of bank credit depends on the individual firm's financial status. However, their bank credit line is not based on an exogenous event, but an endogenous decision that may be determined together with their inventory decisions. Chen et al. (2023) add a causal link between trade credit provision and retail inventory in different sectors. These studies also reveal the importance of studying the heterogeneity in the relationship between financial and operational decisions among different firms. While their work focuses on trade credit, our study investigates how the trade-off between inventory and capacity decisions are affected by bank credit by leveraging the shock due to the interstate-banking deregulation.

Some existing finance studies have provided mixed predictions regarding the role of bank credit on investment. While some suggest that bank credit primarily finances new opportunities (Lins et al. 2010), others propose that undrawn credit primarily serves as bank-monitored liquidity insurance (Acharya et al. 2014). Due to the endogenous relationship between operational decisions and financial availability from banks, many studies utilized financial crises to explore how firms' operational and investment decisions are affected by negative shocks to financial availability. The literature shows that capital investment decreases as credit availability declines during precrisis periods (for example, a 30-cent decrease in investment per 1 euro decrease in credit, based on Cingano et al. 2016). Furthermore, the provision of loans during the peak of a financial crisis often does not lead to increased investments but rather to the accumulation of cash reserves (Acharya et al. 2019). Additionally, the provision of trade credit tends to increase in the short term following a crisis (substitution effect) but contracts in the long term (complement effect) (Love et al. 2007). Our paper differs from this body of work by focusing on a period marked by a positive financial availability shock. Unlike the transient nature of financial crises, this policy change has been sustained, providing a unique context for analysis. We document that during this period, firms would enter a growth phase due to the increased financial availability, characterized by heightened investments and improved operational efficiency. This growth could be supported by both bank credit and trade credit, which act as complementary financial sources. Our work offers a new perspective on the interplay between financial availability and firm operational decisions.

The banking deregulation, seen as a positive financial availability shock, is shown to affect the banking industry, including lower funding costs of banks (Levine et al. 2021), changing composition of interbank entrants (Wang 2019), and expanded competition in the supply of complex mortgages (Cetorelli and Strahan 2006, Di Maggio et al. 2019, Girotti and Salvadè 2022). Moreover, previous research also reveals effects of the banking deregulation on firm activities, including more borrowing of small firms (Rice and Strahan 2010), greater innovation activities of firms (Amore et al. 2013, Cornaggia et al. 2015), corporate risk and volatility (Jiang et al. 2020, Bens et al. 2023), social responsibility investment (Liu et al. 2024), and increased employment and capital growth of young and proactive firms (Bai et al. 2018). Some studies extended this line of study to supply chain networks. Specifically, Shenoy and Williams

(2017) examine how the deregulation affects trade credit in the supplier-customer relationships of U.S. public firms. They find an increase in trade credit from suppliers to customers when the supplier firms have greater access to banking liquidity. However, these studies have generally assumed a homogeneous effect of the regulation on firms' access to bank financing. In contrast, our research leverages the most comprehensive data set on credit lines we obtained through text mining (with details described in Section 3.3.1), allowing us to identify heterogeneous effect on credit line due to this regulation. This approach enables us to identify which types of firms benefit from increased access to credit and to more precisely assess how credit availability and usage influence firms' operational and investment decisions.

2.2. Hypotheses

In exploring the impact of easing bank credit constraints on a company's inventory decisions, we take into account *how and why* a firm's inventory is influenced by its access to the bank credit market. Essentially, when a firm has easier access to bank credit, it can secure funds more readily and at lower costs. Consequently, the firm naturally adjusts its operational decisions to align with these favorable financial changes.

2.2.1. Inventory and Credit Accessibility. With the benefit of improved access to banking liquidity, firms experience an augmentation in their financial positions. This increase in financial health leads to a reduction in the cost associated with acquiring external financing, often referred to as the premium on external finance (Nippani and Green Jr 2002). Consequently, firms are incentivized to accumulate more assets and increase spending (Beck et al. 2005).

In terms of inventory management, the effect of credit accessibility is less straightforward. One might initially expect firms to increase inventory when financial constraints are lifted. Earliest studies by Carpenter et al. (1994) and Kashyap et al. (1994), have shown that when firms grapple with sudden liquidity crunches, they are more likely to curtail their inventory investment compared with other forms of investment, such as fixed assets or research and development, which have higher adjustment costs and are less flexible. In contrast, our research examines a positive shock characterized by an alleviation of financial constraints without a preceding crisis, presenting a different perspective. Do firms' responses to credit easing and tightening occur in a symmetric way? If the response is symmetrical, the firm would increase inventory stocking. Indeed, with greater access to financial resources, one logical reaction could be to expand inventory holdings to better satisfy customer demand, given the relatively low adjustment costs of inventories making them an attractive choice for rapid scaling.

However, firms may have different considerations when credit lines loosen compared with when they tighten, which may lead to asymmetric responses. Given the inherent flexibility of inventory management, firms may be confident in their ability to restock quickly and may prefer to maintain lean inventories rather than holding excess as a precaution. In addition, unlike the transient nature of financial crises, this policy change is expected to be sustained. Therefore, with easier access to bank credit, firms may opt for building more capacity, which can be utilized to respond quickly to demand variety and volume changes and enhance operational performance. Prior theoretical and empirical research consistently demonstrates that a firm's ability to secure external financing is positively correlated with its capacity investment, particularly when firms have imperfect access to credit (Almeida and Campello 2007). The investment in capacity enables a reduction of inventory, especially when firms are faced with unpredictable demand and high obsolescence risk (Raman and Kim 2002). We will further elaborate on this aggregate planning perspective in the next subsection.

In light of these two opposing views, we propose the following competing hypotheses:

Hypothesis 1a. *Increased accessibility to bank credit is associated with lower inventory turnover.*

Hypothesis 1b. *Increased accessibility to bank credit is associated with higher inventory turnover.*

2.2.2. Mechanisms Based on Aggregate Planning Theory. Aggregate planning theory is centered on optimizing the balance between production and inventory over both short- and long-term horizons to meet fluctuating demand while minimizing costs (Mellichamp and Love 1978, Kamien and Li 1990, Vollmann et al. 2004). Key components of this approach include managing capacity, controlling inventory levels, and aligning production schedules with demand variability. Firms decide how much to produce and how much inventory to maintain to meet customer demand efficiently while avoiding the costs associated with overproduction or holding excessive inventory.

In our research context, aggregate planning theory offers valuable insights into how increased access to bank credit influences firms' inventory management and capacity decisions. According to the theory, firms maintain safety stock to mitigate risks of stockouts resulting from fluctuations in demand or supply uncertainties. However, when firms have greater access to bank credit, they gain a new form of financial flexibility. Instead of relying heavily on large inventories to manage demand uncertainty, as well as cycle stocks to offset set-up costs, firms with access to credit can choose to expand their production capacity as they can use bank credit, or leverage undrawn credit as an insurance, to undertake capital investments more easily (Acharya et al. 2014). Limiting inventory levels also helps firms focus financial resources on capacity investment. In the short term, this increased access to credit leads to greater capital investment, accompanied with a reduction in inventory levels.

Over the long term, these capital investments translate into enhanced capacity, such as new equipment, expanded facilities, or additional labor. Increased capacity also allows more frequent set-ups because the loss of production time is offset by higher capacity. This allows firms to meet demand more efficiently, reducing the need for safety stock and cycle stock and decreasing their dependence on costly inventory. With greater production capacity and higher capabilities, firms can produce closer to demand levels, adopting just-in-time production strategies that make operations leaner and more cost-effective. Enhanced capacity provides flexibility, allowing firms to respond more dynamically to market changes, thereby minimizing the risk of stockouts and avoiding costly production adjustments. This improvement in production efficiency leads to more effective inventory management and higher inventory turnover rates.

In summary, increased access to bank credit drives both short-term capital investment and long-term capacity improvements, leading to more efficient inventory management and higher inventory turnover. Therefore, we propose the following hypothesis:

Hypothesis 2. *The relationship between the bank credit accessibility and inventory turnover is mediated by* (a) *capital investment in the short term and* (b) *capital intensity in the long term.*

A firm's inventory and broader operational decisions are not solely determined by its own access to bank credit; the credit access of both its customers and competitors can also play a significant role. Customers with greater access to bank financing may change their purchasing behavior (Almeida and Campello 2007), affecting the firm's inventory turnover and financial planning (Sufi 2009, Demiroglu and James 2011, Wu et al. 2019). Similarly, competitors' access to bank credit could impact market dynamics (Uzzi and Gillespie 2002), influencing how the firm manages its investment, supply chain responsiveness, and profitability (Simerly and Li 2000, Koçak and Özcan 2013, Bustamante and Frésard 2021). These external factors, alongside the firm's own credit situation, create a complex environment where operational and financial decisions must be carefully calibrated to maintain competitiveness and efficiency.

Building on our discussion above, we examines how inventory turnover is influenced by both the focal firms' and their associated external parties' access to bank lines of credit. In the next section, we describe the empirical setting in which we test these hypotheses.

3. Empirical Setting

In this section, we first describe the U.S. banking deregulation that relaxes the constraints on firms access to bank credit. Next, we provide details of our empirical strategy that leverages this policy intervention to identify the impact of banking credit on inventory decisions.

3.1. Background of U.S. Banking Deregulation and IBBEA

For much of the 20th century, U.S. states prohibited banks headquartered in other states from establishing branches within their borders.[2] These regulatory restrictions protected banks from competition, allowed banks to earn monopolistic rents, and therefore created a powerful constituency for maintaining restrictions on interstate banking. Before 1994, banks needed the target state's explicit approval to open branches across state lines. For instance, Maine was the first state to relax its interstate-banking restrictions by enacting a national reciprocal policy in 1978, but no state reciprocated until 1982, when New York adopted a similar nationwide reciprocal agreement and Alaska implemented a national nonreciprocal policy (see Kane 1996). In 1994, the Riegle-Neal Interstate Banking and Branching Efficiency Act (IBBEA) was signed into law. The act removed the remaining obstacles to banks opening branches in other states and provided a uniform set of rules regarding banking in each state (Medley 2013).

The approval of IBBEA was a watershed event for interstate banking but did not immediately lead to nationwide branching in all states. The law permitted states to erect four types of barriers to interstate branching from other states (Rice and Strahan 2010); namely, states could (a) require a minimum age of the acquired institution, (b) restrict interstate branching, (c) disallow the acquisition of individual branches without acquiring the entire bank, and (d) impose statewide deposit caps (see details in Appendix A). Over time, as deregulation of interstate banking progressed after 1994, these barriers were gradually relaxed, allowing banks to expand more freely across state lines and significantly increasing the presence of out-of-state branches nationwide. Before 1994, the average number of out-of-state branches per state was 1.22, with out-of-state branches comprising just 0.0074% of total branches. By the year 2000, the average number of out-of-state branches per state had risen to 323.47, and the percentage of out-of-state branches had increased to 26.03% (Johnson and Rice 2008). Wang (2019) also provides county-level data showing a similar trend—from a very low base level, the number of out-of-state branches increased dramatically following the adoption of deregulation. This clearly indicates that the barriers were relaxed post-IBBEA enactment.

In this study, we interpret the staggered state-level implementation of the IBBEA followed by the relaxation of these barriers as a shock to financial constraints exogenous to individual firms' operations. In the remainder of this section, we describe our data sources in Section 3.2 and variable definitions in Section 3.3, followed by our identification strategy in Section 3.4.

3.2. Data Sources

To investigate how the relaxation of firm financial constraints affects operational decisions, we obtain data and construct variables from the following sources:

1. Compustat Annual Data;
2. Compustat Industry Segment Tape;
3. Institutional Brokers' Estimate System (I/B/E/S);
4. Center for Research in Security Prices (CRSP); and
5. 10-K filings from the Securities and Exchange Commission (SEC).

We obtain firm-specific financial and operational characteristics from the Compustat Annual Database. Considering that existing supply chain network data-sets, such as the FactSet Revere supply chain data set, only contain data starting from 2003, we identify the key customers of firms by using Compustat Industry Segment Tapes to construct the supply chain relationship network for our study. According to the Statement of Financial Accounting Standards (SFAS) No. 14, firms are required to report information for segments, including major customers representing 10% or more of consolidated sales for the fiscal years after 1977. The names of these major customers are given by the Compustat Segment files, which also provide the amount of annual sales to each of them. Each reported major customer name was matched to the company name in the Compustat Annual Files. Because customer names are frequently reported using abbreviations, we first use the same method to Fee and Thomas (2004), then we manually inspect every match to ensure accuracy and manually correct inaccurate and missing matches. We obtain 32,312 supplier-customer links over the study period from 1990 to 2005, averaging approximately 1,800 links per year, consistent with previous literature (Agca et al. 2022).

To compute the cost of capital, we used stock prices in CRSP and analyst forecasts of earnings in I/B/E/S. We collect the data of lines of credit by text-mining firms' 10-K filings from SEC's EDGAR (Electronic Data Gathering, Analysis, and Retrieval) system. In the 10-K filing, a firm typically details the existence of a line of credit and its availability.[3] We follow previous literature (Sufi 2009) to extract the lines of credit information, with details described in Section 3.3.1.

Because the initial introduction of the IBBEA deregulation happened over 1995 to 2001, we select the study period from 1990 to 2005 to ensure it covers the event period. Our sample begins with a Compustat universe that contains nonfinancial U.S.-based firms with at least two years[4] of supply chain information including U.S. customers and positive data on total assets before and after the introduction of IBBEA in the firm's headquarters state. Because our focus is on the inventory and capacity decisions, we exclude the financial service segment.[5] These selections result in a sample of 662 U.S. publicly traded firms, totaling 6,380 firm-year observations.

3.3. Variable Definitions

We construct a set of firm-specific variables based on the unique data set compiled from the sources described above. The main variables are summarized in Table 1.

3.3.1. Dependent Variables: Lines of Credit and Inventory Turnover. Now, we describe the procedures used to construct the lines of credit (see more details in Appendix B). We first follow Sufi (2009) to define a series of terms that indicate lines of credit: "credit lines", "credit facility", "revolving credit agreement", "bank credit line", "working capital facility", "lines of credit", and "line of credit". Then, with the filings that contain the defined terms, we extract the content surrounding the key words[6] and manually collect detailed data by reading the extracted content. It is important to emphasize that, compared with directly reading the entire 10-K filings as in Sufi (2009), we extract the meaningful paragraphs that contain (1) at least one of the defined terms and (2) a dollar sign to reflect an amount. In this way, we collect the data on the amount of credit that a firm can access and the *used* and *unused* proportion of the line of credit. We validate our accuracy by comparing to previous literature (i.e., Sufi 2009), which has a different study time period from 1996 to 2003, and find consistent values. With the extracted data, we use the ratio of *total lines of credit* to current total liabilities (Compustat variable: *LCT*) for the empirical analysis. We also separately consider the *used* and *unused lines of credit* in an additional analysis. To examine the shift in bank loans following the deregulation, we further construct a measurement of short-term debt by calculating the ratio of debt in current liabilities (Compustat variable: *DLC*) to the total assets.

We measure a firm's inventory turnover using the ratio of cost of goods sold (Compustat variable: *COGS*) to ending inventory (Compustat variable: *INVT*) (e.g., Hendricks and Singhal 2009). Our results are robust to an alternative definition of inventory turnover, defined as the ratio of a firm's cost of goods sold to

Table 1. Summary Statistics

Variable	Description	Mean	Median	Std. Dev.	# Obs.
Total Lines of Credit	Total lines of credit scaled by current total liabilities	0.999	0.631	1.578	4,010
Unused Lines of Credit	Unused lines of credit scaled by current total liabilities	0.749	0.413	1.443	4,010
Used Lines of Credit	Used lines of credit scaled by current total liabilities	0.250	0.001	0.458	4,010
Short-Term Debt	Ratio of short-term debt to total assets	0.075	0.021	0.287	6,308
Inventory Turnover	Ratio of cost of goods sold to total inventory	10.523	4.875	21.039	6,308
IBBEA	Indicator that equals to 1 if IBBEA deregulation progressed	0.508	1	0.500	6,308
Customer Relax	Indicator that equals to 1 if deregulation progressed for at least one customer	0.522	1	0.500	6,308
Capital Investment	Ratio of capital expenditure to total assets	0.104	0.078	0.096	6,308
Capital Intensity	Ratio of total property, plant, and equipment to total assets	0.496	0.500	0.221	6,308
Firm Size	Logarithm of total assets	4.972	4.808	1.967	6,308
Cost of Capital	Weighted average cost of capital (WACC)	0.095	0.071	0.104	6,308
Gross Margin	Net sales less the cost of goods sold	0.221	0.308	0.883	6,308
Sales Surprise	Actual-forecast ratio of earnings per share	0.609	0.608	0.784	6,308
Trade Credit	Ratio of accounts payable to current total liabilities	0.395	0.376	0.185	6,308
Cash Flow	Sum of income before extraordinary items and Depreciation and Amortization scaled by total assets	0.021	0.079	0.264	6,308

Notes. There are missing values in lines of credit due to the U.S. SEC's EDGAR system being available since 1994 (see details at https://www.sec.gov/os/accessing-edgar-data). We exclude the missing values from our analysis.

its average inventory in the same year (e.g., Gaur et al. 2005).

3.3.2. Key Independent Variable: Deregulation Implementation. As noted in our empirical strategy, we leverage the IBBEA deregulation as an exogenous shock to lines of credit. Following previous finance and economics literature (e.g., Amore et al. 2013, Jiang et al. 2020), we construct an indicator, *IBBEA*, which takes on the value of 1 for a firm-year pair if the firm's headquarters state had at least one of the four regulations relaxed[7]; the variable retains the value of 1 for subsequent years. We also define two other dummy indicators to replace the variable *IBBEA* in order to separately investigate the effects of IBBEA deregulation in the short and long terms. In particular, we use the two-year threshold and define short- and long-term deregulation indicators if the given year is within two years (*Short*) or beyond two years (*Long*) after the deregulation. We perform robustness checks by using the IBBEA index (defined by the number of barriers removed, as per Johnson and Rice 2008) as a continuous treatment variable, while also examining the differential effects of relaxing each of the four barriers as separate treatments (see Appendix E.8).

To explore the effect of easier access through that firm's main customers, we also need an indicator (*Customer Relax*) to account for the deregulation process among customers of the focal firm. We define this indicator to equal 1 if as least one of the major customer firms has had the deregulation progressed in the given year.

3.3.3. Other Independent Variables. Certain factors, such as gross margin, may also be impacted by deregulation and, in turn, influence inventory turnover. To avoid potential omitted variable bias in estimating the effect of deregulation, we must control for these variables. By including common factors documented to be associated with inventory turnover, we can isolate the specific effect of the exogenous shock by accounting for these influences. We follow recent empirical literature (Gaur et al. 2005, Wu et al. 2019, Yang et al. 2021, Chen et al. 2023) to include these control factors in our empirical framework. We control for *firm size*, measured by logarithm of total assets (Compustat variable: *AT*). In addition to firm size, a wealth of other characteristics could potentially be associated with a firm's inventory decisions. *Cash flow* is calculated as the income before extraordinary items (Compustat variable: *IBC*), plus depreciation and amortization (Compustat variable: *DP*), divided by total assets (Li 2021). *Gross margin* is measured by the net sales less the cost of goods sold ($(SALE - COGS)/COGS$). *Sales surprise* is measured by the actual-forecast ratio of earnings per share with data collected from I/B/E/S. Following Chod et al. (2019), we use accounts payable (Compustat variable: *AP*) scaled by current total liabilities to measure *trade credit* provided to focal firms. *Cost of capital* is measured by a modified version of the weighted average cost of capital (WACC) (Wu et al. 2019). The cost of capital of a firm depends on its capital structure, which can be measured using its leverage, denoted as l: the ratio of the total debt to the sum of debt and equity. If the firm pays r_b for every unit of debt and pays r_e (Gebhardt et al. 2001) for every unit of equity per year, then the weighted average of the cost of equity and cost of debt is $WACC = l \times r_b + (1 - l) \times r_e$.

Further, we follow literature (Gaur et al. 2005, Wu et al. 2019, Bae et al. 2022) to construct two additional variables to reflect potential mediating factors through which the effect of lines of credit on inventory manifests. We measure *capital intensity* using the ratio of the value of the fixed assets (Compustat variable: *PPEGT*) to the total assets, and we measure *capital investment* using the ratio of the value of capital expenditure (Compustat variable: *CAPX*) to the total assets.

3.4. Identification Strategy

The primary empirical challenge we encounter in our study is to identify the impact of the ease of bank credit on inventory decisions. This is a complex task due to the inherent simultaneity between financial and operations decisions. We overcome this challenge by exploiting a policy intervention that imparted an exogenous shock to banking liquidity. In particular, we rely on the timing when at least one out of the four restrictions described above is relaxed (as shown in Table A2). We compare within firms before and after the relaxation in the state where the firms are headquartered, and across firms headquartered in states that deregulated or not. In a robustness check, we also examine the status of firms' subsidiaries and find that the results remain consistent when considering the subsidiaries' relaxation. (see Appendix E.4).

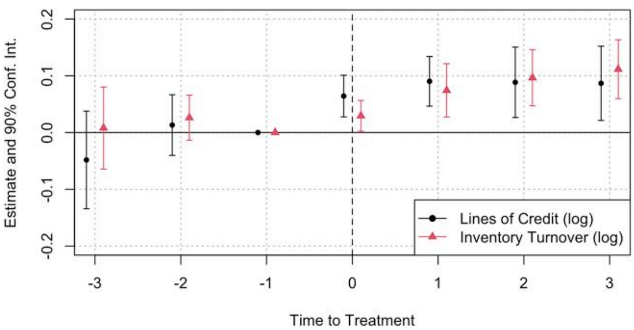
3.4.1. Stacked Difference-in-Differences. While the staggered relaxation of branching restrictions following IBBEA deregulation facilitates a *difference-in-differences (DID)* design (e.g., Beck et al. 2010, Manski and Pepper 2018, Chen et al. 2021, Lin et al. 2021) that isolates the effect of the deregulation on banking credit provisioning and resultant inventory decisions, recent advances in econometric methods suggest that such staggered DID estimates in a simple two-way fixed effects (TWFE) regression are subject to bias due to time-varying treatments, especially when earlier treated units serve as controls for later treated units (Baker et al. 2022). To alleviate this potential bias, we follow recent literature (Cengiz et al. 2019, Slutzky and Zeume 2023, Li 2024) to adopt a stacked regression approach. The main idea of this approach is to create event-specific two-by-two data sets that include the outcome of interest and controls for the treated group and a "clean" control group. Specifically, at the firm level, for each sample year, we create a cohort including first-treated firms (treated) and all firms that are not yet treated for other years (control). We stack all cohorts and implement a regression specification with cohort-specific firm and year fixed effects as follows[8]:

$$\begin{aligned} log(Y_{cit}) &= \beta \cdot IBBEA_{cit} + \gamma \cdot log(Ctrl_{cit-1}) \\ &\quad + Firm\_Cohort_{ci} + Year\_Cohort_{ct} + \epsilon_{cit} \end{aligned} \tag{1}$$

Specifically, we regress the dependent variables for firm i with cohort c in year t, as denoted by Y_{cit}, on the deregulation indicator, as denoted by $IBBEA_{cit}$. β is our coefficient of interest that capture the impact of the banking deregulation for the focal firm. The matrix $Ctrl_{cit}$ contains the various control variables as defined above. We capture the cohort-specific fixed effects by *Firm_Cohort* to account for unobserved firm characteristics and by *Year_Cohort* to adjust for unobserved temporal trends across cohorts. We further cluster the standard errors at the state level to control for correlations in the error terms ϵ_{cit}[9] due to the state-level policy change (Abadie et al. 2023). We apply this specification to all analyses for which we follow the recommendations in Baker et al. (2022). We further confirm that our results from this base model are consistent when using alternative estimators and additional tests to account for the heterogeneity in treatment timing (see Appendix E.2).

3.4.2. Parallel Trends Assumption. To examine the suitability of the DID model, we evaluate the validity of the parallel trend assumption through a set of comprehensive pretrend statistical tests. First, we follow Goodman-Bacon (2021) to perform event-study checks that incorporate covariate controls and a time index, *Time to Treat*, which captures the relative distance in years from the shock to the current time. We then regress both lines of credit and inventory turnover on the relative year indicators. As shown in Figure 1, we observe insignificant coefficients for the years right before IBBEA for both lines of credit and inventory turnover, suggesting that the parallel trends assumption is satisfied. This observation supports the validity of our DID model setup. It implies that the effect of IBBEA deregulation on firm lines of credit and inventory is unlikely to be a result of events that occurred before the policy. Furthermore, the positive and significant estimates for lines of credit for in years after the deregulation provide initial evidence that firms with greater access to banking liquidity could experience an immediate increase in their lines of credit. This finding aligns previous research on IBBEA (e.g., Amore et al. 2013, Keil and Müller 2020), suggesting that the deregulation, which indeed has an immediate effect on banks and borrowers, is a reliable event that reflects relaxing financial constraints of firms. We also note an overall increase in inventory turnover, providing preliminary evidence of the upward shifts in inventory turnover as a result of greater access to bank credit.

Figure 1. (Color online) Per-Year Changes of Bank Credit and Inventory Turnover Before and After Deregulation



Notes. The figure shows how lines of credit and inventory turnover of firms vary in the time periods leading up to and after the deregulation implementation. IBBEA-1 is treated as a baseline and omitted from the estimation (Freyaldenhoven et al. 2021).

In addition, we employ the latest methods—the pretrend equivalence test by Dette and Schumann (2024) and the Honest DID approach by Rambachan and Roth (2023)—to further validate the parallel trends assumption. The results indicate that the estimated effects of IBBEA deregulation on inventory turnover and lines of credit remain significant, even under the maximum pretrend deviations (see Appendix E.1 for details). Moreover, we conduct a series of tests to account for various factors that could potentially influence states' adoption of the deregulation. These tests consider external factors such as regional economic conditions, political influences, and industry-specific trends that may affect when and where the policy was implemented (see Appendix E.3).

4. Analysis and Results

In this section, we first present preliminary checks for the shift in bank credit following the IBBEA deregulation. We then present our main findings on inventory turnover, with a particular focus on distinguishing the effects on firms with and without credit expansions. Next, we examine the potential mechanisms through which focal firms are influenced by the deregulation, particularly through the capital investment. In the post hoc analysis, we further investigate heterogeneity among firms of varying sizes and across different levels of market competition, both directly influenced by the deregulation and indirectly through the supply chain. We conclude the section by offering a series of checks to demonstrate the robustness of our results to alternative model and variable specifications.

4.1. Preliminary Checks for Bank Credit

We perform preliminary checks for testing shifts in firms' lines of credit before and after the deregulation using our staggered DID framework. The estimation results are presented in Table 2. Column (1) shows the overall impact of the deregulation on firms' total lines of credit. The positive and significant coefficient of *IBBEA* ($\hat{\beta} = 0.086$, $p < 0.05$) suggests that total lines of credit out of current liability increases by 8.6% after the deregulation.

We perform an additional analysis to separately check the shift in the allocation of used and unused lines of credit and short-term debt for firms following the banking deregulation. As shown in columns (2) and (4) of Table 2, we find that with the increasing lines of credit, firms keep more credit available and there is no significant change in the used lines of credit after the deregulation. This echoes with the insurance theory (Acharya et al. 2014) that unused bank credit serves as bank-monitored liquidity insurance that facilitates firms' future investments.

Next, we extend our analysis to evaluate the effect of the firm-level expansion of bank credit, differentiating between firms with increased credit and those without. We refer to the following model (2) as the *micro* model because it utilizes individual firm-level credit line data, in contrast to model (1), which only uses IBBEA as a *macro* shock to the system. We employ the following specification:

$$\begin{aligned} log(y_{cit}) = \alpha_1 \cdot IBBEA_{cit} \times Increase_i + \alpha_2 \cdot IBBEA_{cit} \\ \times NonIncrease_i + \gamma \cdot log(Ctrl_{cit-1}) \\ + Firm\_Cohort_{ci} + Year\_Cohort_{ct} + \epsilon_{cit}, \quad (2) \end{aligned}$$

where α_1 and α_2 capture the effects of IBBEA deregulation on inventory turnover conditional for credit-increase (treatment) and nonincrease (control) firms, respectively. As shown in columns (3) and (5) of Table 2, only firms with expanded lines of credit experience a significant increase in unused credit, along with a marginal increase in used credit with a much smaller magnitude. This finding further supports our argument that, following the deregulation, firms with increased credit access tend to utilize some but not all the available bank credit, and retain more available credit as an unused bank credit.

Next, we further examine firms' financing choices in terms of their short-term debt and account payable after the deregulation. Note that both bank credit and trade credit serve as important financing sources for firm investment. Finance literature predicted two possible dynamics for financial choices between these two financial sources, following increased availability of bank credit. Firms could directly use bank debt at a lower cost (Rice and Strahan 2010), or use their available credit as bank-monitored liquidity insurance to lever more trade credit from their suppliers (Acharya et al. 2014). Our result in in column (1) of Table 3 shows that there is a significant increase in short-term debt following the deregulation, especially for firms with credit expansion, suggesting that these firms

Table 2. Effects of Banking Deregulation on Lines of Credit

	Total credit	Unused credit		Used credit	
	(1)	(2)	(3)	(4)	(5)
IBBEA	0.086***	0.093***		−0.002	
	(0.022)	(0.018)		(0.025)	
IBBEA × Credit Non-Increase			0.012		−0.048
			(0.028)		(0.031)
IBBEA × Credit Increase			0.173***		0.043*
			(0.020)		(0.023)
Size	−0.055***	−0.077***	−0.083***	0.019**	0.015*
	(0.015)	(0.018)	(0.018)	(0.009)	(0.008)
Gross Margin	0.065*	0.100**	0.109**	−0.030**	−0.025*
	(0.034)	(0.040)	(0.042)	(0.014)	(0.013)
WACC	−0.011	0.031	0.033	−0.053	−0.052
	(0.052)	(0.062)	(0.061)	(0.043)	(0.043)
Sale Surprise	0.003	−0.002	−0.000	0.006	0.007
	(0.021)	(0.021)	(0.020)	(0.014)	(0.014)
Trade Credit	0.284***	0.389***	0.388***	−0.069	−0.070
	(0.094)	(0.098)	(0.098)	(0.046)	(0.046)
Cash Flow	0.060	0.065	0.071*	−0.002	0.001
	(0.047)	(0.040)	(0.041)	(0.019)	(0.020)
Firm-cohort (FE)	✓	✓	✓	✓	✓
Year-cohort (FE)	✓	✓	✓	✓	✓
Num. obs.	4,010	4,010	4,010	4,010	4,010
R^2 (full model)	0.683	0.641	0.645	0.616	0.619

Notes. Columns (1), (2), and (4) estimate the overall effect of deregulation (macro), while the remaining columns differentiate between firms that experienced credit expansion and those that did not (micro). Column (1) presents estimation with total lines of credit as the dependent variable. Columns (2) and (3) show results with unused lines of credit as the dependent variable, while columns (4) and (5) focus on used lines of credit. Standard errors are presented in parentheses.
***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

lever more bank loans when they have greater access to bank credit. When applying the *micro* approach outlined in Equation 2, as shown in column (2) of Table 3, we find that firms with credit expansion see an increase in short-term debt, suggesting that these firms lever more bank loans when they have greater access to bank credit. Furthermore, to alleviate potential concern on comparability between these groups after the shock, we further apply inverse probability/propensity weighted (IPW) matching (Imbens and Wooldridge 2009), which

Table 3. Effects of Banking Deregulation on Short-Term Debt and Account Payable

	Short-term debt			Account payable		
	Macro (1)	Micro (2)	Micro w/IPW (3)	Macro (4)	Micro (5)	Micro w/IPW (6)
IBBEA	0.011**			0.011**		
	(0.004)			(0.005)		
IBBEA × Credit Non-Increase		−0.007	−0.006		−0.001	−0.001
		(0.006)	(0.007)		(0.007)	(0.007)
IBBEA × Credit Increase		0.013***	0.013***		0.017***	0.020***
		(0.005)	(0.005)		(0.006)	(0.007)
Controls	✓	✓	✓	✓	✓	✓
Firm-cohort (FE)	✓	✓	✓	✓	✓	✓
Year-cohort (FE)	✓	✓	✓	✓	✓	✓
Num. obs.	6,308	4,010	4,010	6,308	4,010	4,010
R^2 (full model)	0.607	0.697	0.711	0.681	0.783	0.783

Notes. Columns (1)–(3) and (4)–(6) report specifications using short-term debt and accounts payable as the dependent variables, respectively. Columns (1) and (4) use macro analysis while the remaining columns represent microlevel results that differentiate between firms with credit increase and those without; columns (3) and (6) utilize a weighted sample to balance firm characteristics between the two groups. For simplicity, control variables are omitted in this and subsequent tables. Standard errors are presented in parentheses.
***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

aims to reduce bias by making the observed covariates more balanced, while making it possible to use all the available observations. This method is widely used to estimate treatment effects in management and economics research (e.g., Zhang et al. 2022, Dimitriadis and Koning 2024). We use this method to create weighted samples that balance the financial conditions, sizes, and operational characteristics between firms with credit expansion and those without in the years prior to deregulation (see Appendix C for details of balancing results). The estimation results using weighted samples are presented in column (3) of Table 3, showing consistent results.

In addition to utilizing bank loans, firms may also use their increased undrawn credit as insurance to secure more trade credit for financing production investments. With both macro and micro approaches (including micro with IPW) applied, as shown in columns (4)–(6) of Table 3, accounts payable (trade credit) rises for firms with expanded bank credit following the deregulation, supporting the insurance theory. Therefore, we observe increased usage from both financing source, supporting predictions from both Rice and Strahan (2010) and Acharya et al. (2014). Moreover, in Appendix E.5, we provide extra evidence consistent with previous literature (Wu et al. 2019), showing that financially constrained firms with low liquidity are more likely to leverage increased lines of credit as an insurance to secure additional trade credit.

In what follows, we explore how the banking deregulation affected inventory turnover among firms and across periods and, furthermore, examine possible mechanisms through which these effects manifest.

4.2. Impact of Banking Deregulation on Inventory Turnover

In this section, we analyze the impact of expanded bank credit on firms' operational decisions using both *macro* and *micro* approaches. By combining these two perspectives, we provide a comprehensive view of how increased credit access influences both broad trends and specific operational decisions within firms.

We start with including only the *IBBEA* indicator together with the fixed effects in the regression model. As reported in column (1) of Table 4, we observe that the estimated coefficient of the variable is positive and significant ($\hat{\beta} = 0.061$, $p < 0.01$). As we control for the year-cohort fixed effects together with firm-cohort fixed effects, this staggered DID scheme indicates that the inventory turnover of firms increases by approximately 6% on average when they have easier access to banking liquidity after the IBBEA deregulation. To control for the effect of operational and financial factors, we include the control variables defined in Section 3.3.3 and continue to observe positive and significant coefficients of *IBBEA*, as shown in column (2) of Table 4. This shows that even with the operational and financial factors controlled, the positive effects are robust and consistent. These results provide evidence supporting Hypothesis 1c.

Next, we employ the micro approach described in Equation 2 to investigate how firm-specific bank credit expansions affect inventory turnover, distinguishing between firms that experienced a credit increase and those that did not. The estimation results for Equation 2 are presented in columns (3) and (4) of Table 4, using both original and weighted (with IPW applied) samples, respectively. The coefficients for the interaction term,

Table 4. Effects of IBBEA Deregulation on Firm Inventory Turnover

	DV: Inventory turnover			
	Macro		Micro	Micro w/IPW
	(1)	(2)	(3)	(4)
IBBEA	0.061^{***}	0.061^{**}		
	(0.022)	(0.023)		
IBBEA × Credit Non-Increase			0.025	0.011
			(0.039)	(0.040)
IBBEA × Credit Increase			0.096^{**}	0.110^{***}
			(0.038)	(0.039)
Controls		✓	✓	✓
Firm-cohort (FE)	✓	✓	✓	✓
Year-cohort (FE)	✓	✓	✓	✓
Num. obs.	6,308	6,308	4,010	4,010
R^2 (full model)	0.852	0.858	0.908	0.900

Notes. Columns (1)–(4) present specifications with inventory turnover as the dependent variable. The first two columns report the macro-level estimated effects of deregulation. The subsequent columns provide microlevel results that differentiate between firms with credit expansion and those without, with Column (4) utilizing a weighted sample to balance firm characteristics between the two groups. Standard errors are presented in parentheses.

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

IBBEA × *Credit Increase*, remain consistently positive and significant across both samples. This suggests that firms, particularly those with increased access to bank credit, experience a notable rise in inventory turnover following the deregulation, with the increase exceeding 9%. These results provide further evidence supporting Hypothesis 1c.

4.3. Mechanism Examination: Short Term vs. Long Term

Our main test result shows that at the individual firm level, the observed results align with an effect resulting from aggregate planning (Kamien and Li 1990). The shock in availability of banking credit, led by the IBBEA deregulation, can potentially influence firms' inventory choices by influencing their operational and investment strategies. Since such a shift of investment consideration is likely to be significant in a short period after the IBBEA deregulation, we extend our analysis to account for both short- and long-term effects of this event using the *micro* approach, which leverages on credit expansion following the banking deregulation.[10]

We check the dynamics of capacity expenditure (*CAPX*) and capital intensity (*CI*). In column (1) of Table 5, we observe a significant increase in capacity investment in the short term, while column (2) shows that higher banking liquidity positively affected capital intensity for credit-increase firms and that the effect is both statistically and economically stronger in the long term.

We then deploy a regression-based procedure of the parallel multimediator model (Hayes 2017) on the inventory equation and present the estimation results in Table 5. As a benchmark, column (3) includes cross terms between short (within two years since deregulation) versus long (more than two years after deregulation) terms and credit-increase versus Nonincrease firms. We observe that the positive effect of the deregulation on credit-increase firms' inventory turn is significant both in the short term ($\hat{\beta} = 0.096$, $p < 0.05$) and in the long term ($\hat{\beta} = 0.123$, $p < 0.05$). This result is consistent with the aggregate planning theory, which emphasizes the substitution effect between capacity investment and inventory investment. Hence, we next test the possible dynamics between the two investments in the long and short terms.

Under the mediation framework (Hayes 2017), an independent variable (here, the IBBEA deregulation) affects a dependent variable (inventory turnover) through the mediators. Specifically, we sequentially add two mediators, capacity investment (*CAPX*) and capital intensity (*CI*), to the model and report estimated coefficients in columns (4)–(6). When we add *CAPX* in column (4), we find that its coefficient is positive. By comparing column (4) to column (3), we observe that, with *CAPX* controlled, the short-term effect of the banking deregulation on inventory turnover diminishes (the estimate of *Short Term* × *Credit Increase*, $\hat{\beta} = 0.074$, $p > 0.1$). This comparison further echoes the aggregated planning theory that the inventory decisions are codetermined by capital investment

Table 5. Mediation Effects of CAPX and CI (Micro: Credit Expansion)

	CAPX	CI	Inventory turnover			
	(1)	(2)	(3)	(4)	(5)	(6)
Short Term × Credit Non-Increase	0.012	0.016*	0.022	0.013	0.001	−0.012
	(0.016)	(0.009)	(0.033)	(0.039)	(0.030)	(0.036)
Short Term × Credit Increase	0.028**	0.011	0.096**	0.074	0.082**	0.054
	(0.013)	(0.009)	(0.045)	(0.053)	(0.041)	(0.048)
Long Term × Credit Non-Increase	0.014	0.019	0.052	0.042	0.028	0.012
	(0.019)	(0.013)	(0.048)	(0.056)	(0.045)	(0.052)
Long Term × Credit Increase	0.010	0.037***	0.123***	0.115**	0.075	0.060
	(0.020)	(0.012)	(0.044)	(0.054)	(0.045)	(0.055)
CAPX				0.773***		0.938***
				(0.142)		(0.142)
CI					1.290***	1.426***
					(0.193)	(0.182)
Controls	✓	✓	✓	✓	✓	✓
Firm-cohort (FE)	✓	✓	✓	✓	✓	✓
Year-cohort (FE)	✓	✓	✓	✓	✓	✓
Num. obs.	4,010	4,010	4,010	4,010	4,010	4,010
R^2 (full model)	0.583	0.887	0.908	0.910	0.914	0.918

Notes. Columns (1) and (2) show specifications where the dependent variable is capital investment and capital intensity, respectively; columns (3)–(6) show specifications where the dependent variable is inventory turnover. For each dependent variable, we examine the effect of the deregulation in different time periods and compare firms with credit expansion to those without. Standard errors are presented in parentheses.

***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

in the short term. Similarly, in column (5), we account for capital intensity and find that it has a positive and significant association with inventory turnover, which echoes the presumption that higher capital intensity increases inventory turnover (Gaur et al. 2005). In addition, adding capital intensity, we find that the long-term effect of the deregulation on inventory is eliminated (the estimate of *Long Term* × *Credit Increase*: $\hat{\beta} = 0.075, p > 0.1$) while the short-term estimate remains significant for credit-increase firms. This comparison suggests that the effects of banking liquidity on inventory turnover manifest through capital intensity in the long term. Following Preacher and Hayes (2008), we estimate that the total indirect effect of the deregulation on inventory turnover through the mediators (CAPX and CI), based on column (6), is 0.080 ($p < 0.01$[11]). We further validated the significance level of the indirect effects using a bootstrapping analysis (Preacher and Hayes 2008).

In summary, our findings indicate that increased capital investment enhances firms' capacity to meet demand, allowing them to operate more efficiently by reducing their reliance on safety stock, which is typically used as a buffer against demand or supply uncertainties. This leads to higher inventory turnover, reflecting more effective inventory management. This result provides strong support for Hypothesis 4.

4.4. Post-Hoc Analysis: Firm and Market Heterogeneity

In this section, we discuss how banking deregulation impacts bank credit access and firm inventory management in a boarder supply chain context, focusing on the role of competition as a key factor. Competition affects a firm's cash flow risk and default likelihood, and, consequently, the level of competition in the product market often influences both the cost and availability of its bank credit (Hasan et al. 2021). By considering the level of market concentration across industries and firms' position within markets, we explore how these competitive dynamics influence the relationship between deregulation, credit access, and firms' inventory management strategies. Due to space limitations, detailed posthoc analysis results are provided in Appendix D; we summarize the key insights herein.

4.4.1. Effects of Competition. We find that market concentration, as measured by the Herfindahl-Hirschman Index (HHI), significantly affects firms' access to bank credit following deregulation (as shown in Table D1). Specifically, firms in concentrated markets, characterized by higher HHI, experience increased access to bank credit following the deregulation, likely due to their stable profit margins, secure market shares, and lower financial risk (Tremblay and Tremblay 2012). Conversely, firms in competitive markets show no significant change in credit accessibility postderegulation, highlighting that market dynamics distinctly shape how firms benefit from eased banking constraints.

In addition to market concentration,[12] individual firm characteristics, such as firm size, can also influence credit accessibility, even in the same market. Defining firm size based the industry median of firms' total assets, we find that smaller firms within concentrated markets notably has improved access to bank credit, underscoring an asymmetric impact driven by firm size and market positioning, as banks anticipate greater relative benefits for smaller firms once they have increased financial resources. This greater access to credit has improved efficiency among smaller firms, as they have significantly higher inventory turnover. In contrast, larger firms, which are more likely to hold dominant market positions in concentrated markets, may experience downward pressure on their gross margins over the long term. Taken together, our findings suggest that the benefits of deregulation and increased competition are not evenly distributed among firms with different market positions: smaller firms in concentrated markets experience greater gains, while larger firms may face reduced profitability due to intensified competition.

4.4.2. Indirect Effects from Major Customers. In a broader supply chain context, we find that firms' inventory turnover is positively influenced when their major customers gain improved access to bank credit (as shown in Table D2). This effect is especially prominent among larger supplier firms operating in competitive markets, primarily due to increased sales volumes driven by customers' enhanced purchasing capabilities. we also observe that large firms in competitive markets see an increase in their accounts receivable after their customers have greater access to bank credit, indicating that these firms extended more trade credit to their customers when their customers had greater access to bank credit. These findings together underscore the interconnectedness of financial and operational decisions across the supply chain, indicating that increased financial capacity at the customer level effectively cascades upward, benefiting supplier operational performance. Thus, enhanced bank credit not only influences individual firms but also significantly impacts inventory management strategies and operational efficiencies across related supply chain partners.

Overall, our findings present the first evidence that the relaxation of financial constraints through improved bank credit accessibility is associated with increased inventory turnover. Importantly, these positive effects stem from both the firm's own access to banking resources and that of its customers, with the self-effect being more significant for small firms in concentrated markets, while the customer effect is stronger for larger firms in competitive markets. In Section 5, we discuss our findings and offer theoretical and practical implications.

4.5. Assumption Validity and Robustness Analysis

We perform a series of extra examinations to confirm that our main analysis framework is valid and our main results are robust to alternative specifications and variables. We summarize our robustness analyses in Table 6 (see tables and additional details in Appendix E).

5. Discussion and Conclusion

Our study provides a nuanced view of how credit accessibility affects firms' financial and operational decisions, market competition, and supply chain relationships. Our findings provide valuable insights for managers aligning financial strategies with operations

Table 6. Assumption Validity and Robustness Analysis

Potential concern	Empirical examination	Appendix
	Panel A: Model assumption	
Parallel Trends Assumption	(1) We test the equivalence of pretrends by focusing on a subsample of preperiods and show that any potential violations of the parallel trends assumption are minimal.	E.1.1
	(2) We perform the Honest DID analysis and demonstrate that even as allowances for violations widen confidence intervals, the observed impacts remain significant.	E.1.2
	(3) We perform falsification tests by randomly assigning placebo policy dates and find that the placebo effects are mostly insignificant, validating the model assumption.	E.1.3
Alternative Estimators	We ensure the robustness of our stacked model using alternative staggered DID estimators: (1) Callaway and Sant'Anna DID framework, which confirms no pretreatment effects and supports our main findings, (2) Goodman-Bacon decomposition, which shows consistent treatment effects, and (3) Wing's weighted stacked approach to address stack weights.	E.2
	Panel B: Alternative explanations	
Deregulation Timing	We compare early- and late-adopting states by interacting the subgroup indicator with the treatment and show both groups exhibit similar effects, suggesting that our results are not likely biased by timing differences.	E.3.1
State Economic and Demographic Factors	We confirm the robustness of our results after controlling for state-level economic and demand factors, measured annually using multiple indicators: (1) population, (2) GDP, (3) personal income, and (4) energy consumption. Additionally, we ensure consistency by accounting for (5) the state-level banking market reflected in the fraction of small banks, and (6) the political environment.	E.3.2
	Panel C: Additional heterogeneity factors	
Firm-Level Factors	(1) We account for firm subsidiaries in states different from their headquarters and show an insignificant effect, confirming that our findings remain robust across firms with subsidiaries.	E.4
	(2) We examine firm financial liquidity and show prederegulation Z-scores and trade credit levels were similar between firms with and without credit expansion, suggesting no prior financial differences. We also confirm that nonliquid firms experienced greater inventory turnover increases, indicating a stronger reliance on bank credit.	E.5
Industry-Level Factors	We examine how (1) capital intensity and (2) demand volatility influence industry responses to deregulation. Firms in capital-intensive industries are more likely to invest in capacity, while those in volatile-demand industries adjust inventory management, confirming our proposed link between credit access and investment behavior.	E.6
	Panel D: Additional robustness checks	
Alternative Specifications	We confirm the robustness of mechanism analysis by employing alternative specifications: (1) a macro-level analysis and (2) a microlevel analysis using inverse probability weighting (IPW).	E.7
Alternative Variables	(1) We use a continuous deregulation index based on the number of removed barriers, finding consistent results. Analyzing barriers separately, all except the "Statewide Deposit Cap" significantly impact credit and inventory, suggesting it is less restrictive on bank credit supply and firm operations.	E.8
	(2) We redefine credit expansion as a continuous variable based on percentage growth in credit lines and show consistent results.	E.9
Alternative Samples	We repeat our analysis using all nonfinancial firms, regardless of customer data availability, ensuring results are not biased by sample selection nor supplier-driven when requiring supply chain information.	E.10

and for policy-makers considering the impacts of credit accessibility on market dynamics.

5.1. Findings and Implications

Prior to the implementation of IBBEA, both government and academic research (e.g., Savage 1993, Golembe 1994, Kane 1996) primarily focused on the anticipated impact of the law in the banking industry. After the IBBEA's enactment, there are studies on the effects of this deregulation on factors such as firms' innovation, borrowing behavior and employment etc. However, one area that remained unexplored was the impact of bank credit relaxation on firms' inventory decisions. Hence, during the policy-making process, it is likely that effects of this deregulation on inventory management and resulting impact on product market were neither fully anticipated nor well established.

Our study suggests that with a systematic relaxation of financial constraints, particularly in the form of increased access to bank credit, firms do not in general increase their inventory investment. Instead, they adopt a growth-oriented strategy, leveraging the newly available financial resources for capital investments in the short term. Firms increasingly rely on trade credit, using undrawn bank credit as a form of financial insurance, which complements bank loans in supporting capital expenditures. Over the long term, this approach enables firms to manage their operations more efficiently, leading to improved inventory turnover. These findings have important implications for policymakers, helping them understand how firms operationally respond to a sustained positive shock to financial constraints. Our results underscore the overall economic benefits of loosening credit constraints, as it can drive greater firm-level efficiency in inventory management and productivity in general. These insights can inform future regulatory decisions by clarifying the broader economic consequences of financial liberalization and access to capital.

Our study also highlights the heterogeneous effects of financial deregulation across firms in different industries. We find that smaller firms, particularly those in concentrated markets, stand to benefit the most from relaxed credit constraints, as they are able to invest more aggressively and enhance their operational efficiency. This often comes at the expense of larger firms, which may face increased competitive pressure. Therefore, not all firms benefit from deregulation; for example, while smaller firms may thrive, larger firms in concentrated markets may encounter challenges such as declining gross margins in the long term. For managers, these findings offers critical insights into how changes in financial conditions can influence operational decisions and supply chain management. In addition, our paper emphasizes the importance of not only focusing on the impact on the firm's own financial resources but also considering broader market dynamics. Managers of large firms in concentrated markets, in particular, should stay vigilant about shifts in the competitive landscape. Strategic responses, such as investing in innovation, diversifying product lines, or offering more flexible trade credit terms, may help firms maintain their competitive edge and capitalize on new opportunities. A broader vision that includes strategies of other key market players and supply chain partners will better position the firm to navigate the evolving economic environment.

5.2. Limitations and Future Research

We acknowledge several limitations inherent in our analyses. Firstly, our study focuses solely on U.S. public firms, thus excluding private firms and firms from other countries. This limitation restricts the generalizability of our findings to a broader context. In addition, due to data limitations, we are only able to identify the main customers associated with the firms under investigation. In future research, it would be valuable to access more detailed data, which would enable a customer-centric examination of the mechanisms through which a firm's bank credit accessibility impacts other members in the supply chain. Furthermore, our findings raise questions that relate to classical finance theories, such as pecking order theory, which address firms' prioritization among different financing sources. Future research could further investigate this direction by utilizing more comprehensive data on information availability to each party and the associated relative costs. Lastly, despite employing various empirical techniques and identification strategies to mitigate endogeneity concerns, we acknowledge that we cannot completely address endogeneity, particular stemming from unobserved firm-level heterogeneity, solely using observational data. Identifying solid instrumental variables or conducting an actual field experiment could further enhance the robustness of our findings and provide deeper insights.

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Endnotes

[1] Data source: Compustat database.

[2] The following description of regulatory changes to cross-state banking state and federal policy is summarized mainly from Johnson and Rice (2008) and Rice and Strahan (2010).

[3] Although full credit line details were not always mandatory, GAAP and SEC regulations, such as Regulation S-K, required firms to disclose significant borrowing arrangements, covering liquidity and capital resources. Firms often included credit facility

information in financial statement footnotes, detailing terms such as interest rates, maturity dates, and covenants (Sufi 2009). In our sample, 82.3% of firm-years reported a credit line, and 92% of firms had credit line information at least once in our study period, suggesting minimal impact from any missing reports.

[4] Our results are robust from using the sample with at least *three* or *five* observations before and after the shock.

[5] We filter out the financial service segment by excluding (1) *Industry Format* that equals to "FS" (i.e., Financial Services, includes banks, insurance companies, broker/dealers, real estate, and other financial services) and (2) firms with *SIC code* in the range of 6000 to 6799.

[6] We used the Python package "BeautifulSoup" to extract 10K filings through the SEC's EDGAR system (https://www.sec.gov/search-filings).

[7] As noted, the four barriers set by states include (i) the minimum age of the target institution, (ii) de novo interstate branching, (iii) the acquisition of individual branches, and (iv) a statewide deposit cap.

[8] We use the R package "fect" and tutorial by Yiqing Xu, available at https://yiqingxu.org/tutorials/panel.html.

[9] We verify that our results remain robust when we cluster the error terms at the firm level.

[10] Our results are consistent when employing (1) a macro-level analysis and (2) a micro-level analysis with weighted samples (Appendix E.7).

[11] To determine the significance of this effect, we find that the asymptotic variance is 0.0005. The asymptotic critical ratio for the total indirect effect is therefore $Z = \frac{0.080}{\sqrt{0.0005}} = 3.52$, which leads to a rejection of the null hypothesis that the total indirect effect is zero ($p < 0.01$).

[12] In addition to market concentration, we also explored other industry-specific factors that may influence the effects of bank credit on inventory turnover (e.g., capital intensity and demand volatility), with details described in Appendix E.6.

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